VUNGU RURAL DISTRICT COUNCIL

GIS-Integrated Municipal Planning Information System

FRONTEND SYSTEM DESIGN DOCUMENT

Conceptual Model • Logical Model • Physical Model
Application Windows • User Roles • Report Formats

Document Title	Frontend System Design Document
Organisation	Vungu Rural District Council (Vungu RDC)
System	GIS-Integrated Municipal Planning Information System
Version	1.0
Date	April 2026

1. SYSTEM OVERVIEW

The GIS-Integrated Municipal Planning Information System (hereafter "the System") is a web-based geospatial platform developed for Vungu Rural District Council (Vungu RDC), Gweru Rural, Zimbabwe. The system digitalises and integrates the Council's planning, land use management, development application processing, environmental oversight, and reporting functions into a single, role-aware, map-centric application.

The system serves the Council's five departments and two units: Finance, Planning and Environment, Social Services, Roads and Works, Human Resources and Administration, Internal Audit, and Procurement Management. It also provides a public-facing portal through which landowners, developers, and citizens may submit development applications, track permit status, and access authorised spatial datasets.

1.1 Technology Stack

Layer	Technology	Purpose
Frontend	Vue 3 + Vite + TypeScript	Single-page application, component framework
State Management	Pinia	Global authentication and spatial data stores
Routing	Vue Router	Client-side navigation with role-based guards
Map Engine	MapLibre GL JS	WebGL map rendering, GeoJSON layer display
Spatial Analysis	Turf.js	Client-side area, distance, buffer calculations
Animations	GSAP + ScrollTrigger	Scroll-triggered UI animations
HTTP Client	Axios	JWT-authenticated API communication
Icons	Lucide Vue Next	UI icon library
Styling	Tailwind CSS + Scoped CSS	Utility classes and component styles
Backend API	Node.js + Fastify	RESTful JSON API with OGC proxy
Spatial DB	PostgreSQL + PostGIS	Geometry storage, spatial queries
Map Tiles / OGC	QGIS Server / GeoServer	WMS tile serving, WFS feature serving
Deployment	Nginx + PM2 + Docker	Reverse proxy, process management, containers

1.2 Geographic Scope

The system covers the full spatial extent of Vungu Rural District Council, comprising 20 wards across the Gweru Rural area.

2. SYSTEM PLAYERS — ROLES AND JOB DESCRIPTIONS

The system recognises the following user roles, derived from the Vungu RDC organisational structure. Each role has a defined set of permissions, a primary landing screen, and specific functions within the system.

Role Name	System Role Code	Org. Position	Landing Screen
System Administrator	admin	IT / CEO office	/admin/users
E.O. Planning & Environment	eo	Dept Head — Planning	/eo-planner
Planning Officer	planner	Under E.O. Planning	/planner
Building Inspector	inspector	Under Planning Officer	/planner
Environment Officer	environment	Under E.O. Planning	/eo-planner
Council Treasurer	treasurer	Finance Dept Head	/finance
E.O. Income	finance_income	Under Council Treasurer	/finance/income
E.O. Expenditure	finance_expenditure	Under Council Treasurer	/finance/expenditure
Revenue Clerk	revenue_clerk	Under E.O. Income	/finance/revenue
Debtor's Clerk	debtors_clerk	Under Revenue Clerk	/finance/debtors
Accounts Clerk	accounts_clerk	Under E.O. Expenditure	/finance/accounts
Auditor	auditor	Internal Audit Unit	/audit
Audit Clerk	audit_clerk	Under Auditor	/audit
HR & Admin Officer	hr_admin	HR & Administration Dept	/admin/users
Ass. E.O. Social Services	social_services	Under HR & Admin	/analytics
Salaries Officer	salaries	Under HR & Admin	/analytics
Roads Engineer	roads_engineer	Roads & Works Dept	/planner
Civil Technician	roads_tech	Under Roads Engineer	/planner
CEO	ceo	Chief Executive	/analytics
Public User	user	External / Public	/

2.1 Chief Executive Officer (CEO)

System Role Code: ceo

The CEO is the head of Vungu Rural District Council and bears overall accountability for the Council's administration, service delivery, and statutory compliance. Within the system, the CEO is a read-only executive user who accesses high-level dashboards, ward-level statistics, and consolidated reports prepared by departmental officers. The CEO does not create, edit, or approve spatial data directly.

Permissions

- View executive dashboard (application volumes, ward statistics, approval rates)
- View and download all departmental reports
- View the interactive GIS map in read-only mode
- View user account list (read-only)
- Cannot create or modify any records

Reports Generated / Accessed

- Monthly Executive Summary Report

- Ward Development Progress Report
- Annual Permit Statistics Report

2.2 E.O. Planning and Environment

System Role Code: eo

The Executive Officer (E.O.) Planning and Environment is the head of the Planning and Environment Department. This is the primary decision-making role in the planning workflow. The E.O. receives planning proposals prepared by Planning Officers for final departmental approval before they proceed to Council. The E.O. also oversees environmental compliance, monitors development activity across all 20 wards, and is responsible for the department's statutory reporting obligations to the Ministry of Local Government. Within the system, the E.O. uses the dedicated E.O. Planning Command Centre — a full-screen, map-centric interface with access to satellite EO imagery layers, environmental overlays, and the application approval queue.

Permissions

- Access the E.O. Planning Command Centre (/eo-planner)
- Review and approve or reject planning proposals (stage: awaiting_eo → approved / rejected)
- Review and make decisions on development applications
- View and toggle all GIS layers including environmental overlays
- View satellite EO imagery layers (Sentinel-2 NDVI, True Colour, Thermal)
- Run spatial conflict checks on any ward or proposal geometry
- View and download all planning and environmental reports
- View all wards' planning statistics
- Cannot create new proposals (that is the Planning Officer's role)

Reports Generated / Accessed

- Monthly Planning Department Report
- Proposal Approval Summary
- Environmental Compliance Report
- Development Activity by Ward Report

2.3 Planning Officer

System Role Code: planner

The Planning Officer is responsible for the day-to-day spatial planning operations of the Council. Duties include preparing residential and commercial layout schemes, processing development applications received from the public, conducting spatial analysis (conflict checks, yield calculations), managing planning tasks and deadlines per ward, and preparing proposals for the E.O.'s approval. Within the system, the Planning Officer uses the Vungu Planner — a three-pane GIS workstation with a full MapLibre GL map, spatial draw/measure/buffer tools, a layer panel, a task list, an approval tracker, and a yield analysis table.

Permissions

- Access the Vungu Planning Command Centre (/planner)
- Create, edit, and submit planning proposals (stage: draft → circulating → awaiting_eo)
- Draw polygons, measure distances, and generate buffer zones on the map

- Run yield calculations on drawn layout polygons
- Run spatial conflict checks on proposed layouts
- Review and process development applications in their assigned wards
- Create and update planning tasks
- View and toggle GIS layers (land use, infrastructure, ward boundaries)
- Export yield tables as CSV
- View ward statistics and search geographic locations

Reports Generated / Accessed

- Development Application Status Report
- Planning Task Progress Report
- Ward Yield Analysis Report
- Conflict Check Report (per proposal)
- Quarterly Ward Planning Summary

2.4 Building Inspector

System Role Code: inspector

The Building Inspector conducts on-site inspections of development works to verify compliance with approved plans, building regulations, and planning conditions. Inspection findings are recorded in the system, linked to the relevant development application, and used to update compliance status. The Building Inspector operates within the planner interface but with a focus on the inspection management and compliance tracking modules.

Permissions

- Access the Vungu Planner in inspection mode (/planner)
- View assigned development applications and their approved plans
- Record inspection findings (date, result, observations, photos)
- Update application compliance status (compliant / non-compliant / pending)
- View the GIS map to locate stands and access property information
- Cannot approve or reject planning proposals

Reports Generated / Accessed

- Inspection Report (per application)
- Monthly Inspection Activity Report
- Non-Compliance Summary Report

2.5 Environment Officer

System Role Code: environment

The Environment Officer is responsible for environmental assessment and oversight within the Planning and Environment Department. Key functions include maintaining environmental overlay layers (wetlands, river buffers, protected areas), conducting environmental impact assessments on proposed developments, and ensuring compliance with the Environmental Management Act (Zimbabwe). The Environment Officer uses the E.O. Planning Command Centre with a focus on environmental layers and the EO satellite imagery stack.

Permissions

- Access the E.O. Planning Command Centre (/eo-planner)
- View and update environmental overlay GIS layers
- Run conflict checks against environmental buffers
- View satellite EO imagery (NDVI, Land Cover, Thermal)
- Record environmental assessment findings on proposals
- View all wards' environmental compliance data

Reports Generated / Accessed

- Environmental Compliance Report
- Environmental Impact Assessment Summary
- Wetland and Buffer Zone Status Report

2.6 Council Treasurer

System Role Code: treasurer

The Council Treasurer is the head of the Finance Department and is responsible for all financial management of the Council including revenue collection, expenditure control, and financial reporting. Within the system, the Treasurer accesses financial dashboards linked to development application fees, yield revenue estimates from planning proposals, and debtor accounts.

Permissions

- Access the Finance Dashboard (/finance)
- View all revenue reports and fee collection summaries
- View yield revenue estimates generated by Planning Officers
- View and manage debtor accounts
- View and download all financial reports
- Cannot create or modify planning or spatial data

Reports Generated / Accessed

- Monthly Revenue Collection Report
- Application Fee Collection Report
- Debtor's Ledger Report
- Annual Financial Summary

2.7 Revenue Clerk / E.O. Income

System Role Code: revenue_clerk

The Revenue Clerk records payments received for development application fees, permit issuance, and other Council charges. Payments are linked to specific applications and stands in the system, providing a full audit trail. The E.O. Income supervises the Revenue Clerk and reviews the revenue collection dashboard.

Permissions

- Record payments against development applications
- View the revenue dashboard and daily collection summaries

- Search applications by applicant or stand number
- Print payment receipts
- Cannot modify application status or spatial data

Reports Generated / Accessed

- Daily Revenue Collection Summary
- Outstanding Fees Report

2.8 Debtor's Clerk

System Role Code: debtors_clerk

The Debtor's Clerk manages outstanding accounts and follows up on unpaid development fees and rates. The system provides a debtor management view listing all stands and applications with outstanding balances, aged debt summaries, and the ability to record partial payments and payment arrangements.

Permissions

- View and manage the debtors list
- Record partial payments and payment arrangements
- Generate aged debtor reports
- Cannot modify spatial data or application decisions

Reports Generated / Accessed

- Aged Debtors Report
- Outstanding Fees by Ward Report

2.9 Auditor / Audit Clerk

System Role Code: auditor

The Auditor and Audit Clerk are responsible for the Internal Audit Unit. They perform independent reviews of financial transactions, planning decisions, and system activity. The system provides the Auditor with a read-only audit log view covering all user actions, application status changes, proposal approvals, and payment records.

Permissions

- View the full system audit log (all user actions, timestamps)
- View all development applications, proposals, and decisions (read-only)
- View all financial records (read-only)
- Export audit logs as CSV or PDF
- Cannot create, modify, or delete any record

Reports Generated / Accessed

- System Activity Audit Report
- Application Decision Audit Trail
- Financial Transaction Audit Report

2.10 HR & Admin Officer

System Role Code: hr_admin

The HR and Administration Officer manages staff records, payroll coordination, and general administration. Within the system, this role has access to the user management module to create, modify, and deactivate staff accounts, assign roles, and send invitation emails to new staff members.

Permissions

- Create, edit, and deactivate user accounts
- Assign system roles to staff members
- Send invitation emails to new users
- View staff activity summaries
- Cannot access planning, finance, or GIS modules beyond basic viewing

Reports Generated / Accessed

- User Account Activity Report
- Role Assignment Summary

2.11 Roads Engineer / Civil Technician

System Role Code: roads_engineer

The Roads Engineer and Civil Technician are responsible for roads infrastructure planning and maintenance. Within the system, they have access to the GIS map for viewing road reserve layers, infrastructure overlays, and proposed development layouts that may affect road alignments. The Roads Engineer can flag road reserve conflicts on planning proposals.

Permissions

- View the GIS map and all infrastructure layers (read-only)
- View planning proposals and flag road reserve conflicts
- View ward statistics and infrastructure data
- Cannot create or approve proposals

Reports Generated / Accessed

- Road Reserve Conflict Report
- Infrastructure Layer Status Report

2.12 Public User (Applicant / Citizen)

System Role Code: user

A Public User is any member of the public who accesses the system via the public-facing portal. This includes property owners, developers, architects, legal representatives, and general citizens. Public Users may register a free account to submit development applications, track the status of their applications, and download approved plans. Unauthenticated visitors may view public GIS data and the layer catalogue without registering.

Permissions

- View the public landing page, about page, and layer catalogue (unauthenticated)
- View the public GIS data explorer map (unauthenticated)
- Register a public account

- Submit development applications (authenticated)
- Track the status of submitted applications (authenticated)
- Download approved plans and compliance certificates (authenticated)
- Cannot access any internal planning, finance, or admin modules

Reports Generated / Accessed

- Application Submission Confirmation (auto-generated PDF)
- Application Status Notification (email)
- Compliance Certificate (when issued by Building Inspector)

3. CONCEPTUAL DATA MODEL

The conceptual model defines the high-level entities in the system and the relationships between them, independent of implementation details. It represents the real-world objects that the system must manage.

3.1 Core Entities

Entity	Description
USER	Any person who interacts with the system — staff, officers, or public applicants.
ROLE	A named permission set assigned to users (e.g. planner, eo, admin).
WARD	One of the 20 administrative divisions of Vungu RDC, each with a defined spatial boundary.
STAND	An individual land parcel (lot) within a ward, identified by a stand number and registered geometry.
ZONE	A land use classification (Residential, Commercial, Industrial, Open Space, Agricultural).
DEVELOPMENT_APPLICATION	A formal request by a property owner or developer to carry out development work on a stand.
PROPOSAL	An internal planning layout scheme prepared by a Planning Officer, covering multiple stands.
INSPECTION_RECORD	A record of a Building Inspector's on-site visit to a stand under an active application.
ENVIRONMENTAL_OVERLAY	A spatial layer representing an environmental constraint (wetland, river buffer, protected area).
GIS_LAYER	A map layer available in the system, sourced from GeoServer or PostGIS.
PAYMENT	A financial transaction recording a fee paid against a development application.
NOTIFICATION	A system message sent to a user (approval, rejection, inspection scheduled, etc.).
REPORT	A generated document summarising data for a specific user, period, or geographic area.
AUDIT_LOG	A timestamped record of every create, update, or delete action performed in the system.

3.2 Key Relationships

Relationship	Cardinality	Description
USER — ROLE	Many-to-one	Each user is assigned exactly one role; a role may be held by many users.
USER — WARD	Many-to-one (optional)	A staff user may be assigned to a primary ward; public users are not ward-specific.
WARD — STAND	One-to-many	A ward contains many stands; each stand belongs to exactly one ward.
STAND — ZONE	Many-to-one	Each stand has one land use zone; a zone applies to many stands.
STAND — DEVELOPMENT_APPLICATION	One-to-many	A stand may have multiple applications over time (e.g. change of use, extension).
USER — DEVELOPMENT_APPLICATION	One-to-many	A public user (applicant) may submit many applications.
DEVELOPMENT_APPLICATION	One-to-many	An application may have multiple inspection

— INSPECTION_RECORD		records at different stages.
INSPECTION_RECORD — USER	Many-to-one	Each inspection is conducted by one Building Inspector.
PROPOSAL — WARD	Many-to-one	Each proposal belongs to a ward.
PROPOSAL — USER (creator)	Many-to-one	Each proposal is created by one Planning Officer.
PROPOSAL — STAND	Many-to-many	A proposal covers a set of stands (via a junction entity PROPOSAL_STAND).
PAYMENT — DEVELOPMENT_APPLICATION	Many-to-one	Multiple payments may be recorded against one application (partial payments).
NOTIFICATION — USER	Many-to-one	Notifications are addressed to a specific user.
AUDIT_LOG — USER	Many-to-one	Each audit entry records the user who performed the action.
GIS_LAYER — WARD	Many-to-many (optional)	Layers may have ward-level filters or extents.

4. LOGICAL DATA MODEL

The logical model normalises the conceptual entities into structured tables, defines all attributes and their data types, and specifies primary keys, foreign keys, and constraints. This model is independent of any specific database engine.

4.1 USERS

Stores all system user accounts — staff and public applicants.

Attribute	Data Type	Constraint	Description
id	UUID	PK	Unique user identifier
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email address
password_hash	VARCHAR(255)	NOT NULL	bcrypt-hashed password (cost factor 12)
full_name	VARCHAR(150)	NOT NULL	User's full name
role_id	INTEGER	FK → ROLES	Assigned system role
ward_id	INTEGER	FK → WARDS (opt)	Primary ward assignment (staff only)
phone	VARCHAR(20)	NULL	Contact phone number
status	VARCHAR(20)	NOT NULL	active | suspended | pending_verification
invite_token	VARCHAR(64)	NULL	Token for staff invitation emails
last_login_at	TIMESTAMPTZ	NULL	Timestamp of most recent successful login
created_at	TIMESTAMPTZ	NOT NULL	Account creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL	Last profile update timestamp

4.2 ROLES

Defines all named permission sets.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Role identifier
name	VARCHAR(50)	UNIQUE	Role code: admin, eo, planner, inspector, etc.
label	VARCHAR(100)	NOT NULL	Human-readable label (E.O. Planning and Environment)
permissions	JSONB	NOT NULL	Array of permission strings granted to this role
landing_path	VARCHAR(100)	NOT NULL	Default redirect path after login

4.3 WARDS

The 20 administrative wards of Vungu RDC.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Ward identifier
name	VARCHAR(100)	NOT NULL	Ward name (e.g. Somabhula, Lalapanzi)
ward_number	INTEGER	UNIQUE	Official ward number (1–20)
population	INTEGER	NULL	Estimated ward population
area_sqkm	NUMERIC(10,2)	NULL	Ward area in square kilometres
stand_count	INTEGER	NULL	Total number of registered stands

center_lng	NUMERIC(12,8)	NOT NULL	Longitude of ward centroid
center_lat	NUMERIC(12,8)	NOT NULL	Latitude of ward centroid
geometry	GEOMETRY(MultiPolygon,4326)	NULL	Ward boundary polygon (WGS84)
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp

4.4 STANDS

Individual land parcels within wards.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Stand identifier
ward_id	INTEGER	FK → WARDS	Ward containing this stand
zone_id	INTEGER	FK → ZONES	Current land use zone
stand_number	VARCHAR(30)	NOT NULL	Official stand number
owner_name	VARCHAR(150)	NULL	Registered owner full name
owner_id_number	VARCHAR(30)	NULL	National ID or company reg number
area_sqm	NUMERIC(12,2)	NULL	Stand area in square metres
street_address	VARCHAR(255)	NULL	Physical address
title_deed_ref	VARCHAR(50)	NULL	Title deed reference number
status	VARCHAR(30)	NOT NULL	vacant | occupied | under_development
geometry	GEOMETRY(Polygon,4326)	NULL	Stand boundary polygon (WGS84)
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL	Last update timestamp

4.5 ZONES

Land use zone classifications.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Zone identifier
name	VARCHAR(100)	NOT NULL	Zone name (e.g. Residential Low Density)
type	VARCHAR(50)	NOT NULL	residential | commercial | industrial | open_space | agricultural | institutional
colour_code	VARCHAR(7)	NOT NULL	Hex colour for map rendering (e.g. #4ade80)
description	TEXT	NULL	Permitted uses and restrictions
max_coverage	NUMERIC(5,2)	NULL	Maximum site coverage percentage
min_setback	NUMERIC(6,2)	NULL	Minimum building setback in metres

4.6 DEVELOPMENT_APPLICATIONS

Formal development requests submitted by property owners.

Attribute	Data Type	Constraint	Description
id	VARCHAR(20)	PK	Application reference (e.g. DA-2026-0001)
applicant_id	UUID	FK → USERS	Submitting user
stand_id	INTEGER	FK → STANDS	Stand subject to the application
application_type	VARCHAR(50)	NOT NULL	new_build | extension | change_of_use | subdivision | demolition

description	TEXT	NOT NULL	Applicant's description of proposed works
submitted_at	TIMESTAMPTZ	NOT NULL	Submission timestamp
status	VARCHAR(30)	NOT NULL	submitted | under_review | awaiting_inspection | approved | rejected | withdrawn
decision_notes	TEXT	NULL	Officer's decision rationale
decided_by	UUID	FK → USERS (NULL)	Officer who made the decision
decided_at	TIMESTAMPTZ	NULL	Decision timestamp
fee_required	NUMERIC(10,2)	NOT NULL	Total application fee due
fee_paid	NUMERIC(10,2)	NOT NULL DEFAULT 0	Total amount paid to date
documents_path	TEXT[]	NULL	Array of uploaded document file paths
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL	Last status update timestamp

4.7 PROPOSALS

Internal planning layout schemes prepared by Planning Officers.

Attribute	Data Type	Constraint	Description
id	VARCHAR(20)	PK	Proposal reference (e.g. PRO-2026-001)
title	VARCHAR(200)	NOT NULL	Descriptive title of the proposal
creator_id	UUID	FK → USERS	Planning Officer who created it
ward_id	INTEGER	FK → WARDS	Ward the proposal falls within
stage	VARCHAR(30)	NOT NULL	draft | circulating | awaiting_eo | approved | rejected | billed
geometry	GEOMETRY(Polygon,4326)	NULL	Drawn layout boundary polygon (WGS84)
area_ha	NUMERIC(10,4)	NULL	Calculated area in hectares
stand_count	INTEGER	NULL	Total planned stands in the layout
notes	TEXT	NULL	Planning Officer's notes
approved_by	UUID	FK → USERS (NULL)	E.O. who approved the proposal
approved_at	TIMESTAMPTZ	NULL	Approval timestamp
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL	Last update timestamp

4.8 PROPOSAL_STANDS

Junction table linking proposals to the individual stands they include.

Attribute	Data Type	Constraint	Description
proposal_id	VARCHAR(20)	PK, FK → PROPOSALS	Proposal reference
stand_id	INTEGER	PK, FK → STANDS	Stand included in this proposal
stand_type	VARCHAR(30)	NOT NULL	residential | commercial | industrial | open_space
plot_size_sqm	NUMERIC(10,2)	NULL	Planned stand size
app_fee	NUMERIC(10,2)	NULL	Application fee for this stand type

4.9 INSPECTION_RECORDS

On-site inspection records by Building Inspectors.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Inspection record identifier
application_id	VARCHAR(20)	FK → DEVELOPMENT_APPLICATIONS	Associated application
inspector_id	UUID	FK → USERS	Building Inspector who conducted the visit
scheduled_date	DATE	NOT NULL	Scheduled inspection date
conducted_date	DATE	NULL	Actual inspection date
stage	VARCHAR(50)	NOT NULL	foundation | slab | walling | roofing | completion
result	VARCHAR(30)	NULL	pass | fail | conditional_pass | not_conducted
observations	TEXT	NULL	Inspector's field observations
defects	TEXT	NULL	Listed defects or non-compliance items
photos_path	TEXT[]	NULL	Array of inspection photo file paths
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp

4.10 ENVIRONMENTAL_OVERLAYS

Spatial constraint layers managed by the Environment Officer.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Overlay identifier
name	VARCHAR(100)	NOT NULL	Overlay name (e.g. Gweru River Wetland Buffer)
type	VARCHAR(50)	NOT NULL	wetland | river_buffer | protected_area | road_reserve | heritage
buffer_distance_m	NUMERIC(8,2)	NULL	Buffer zone width in metres (for line/point sources)
geometry	GEOMETRY(MultiPolygon,4326)	NOT NULL	Overlay boundary or buffer polygon
description	TEXT	NULL	Regulatory basis for the overlay
colour_code	VARCHAR(7)	NOT NULL	Hex colour for map display
opacity	NUMERIC(3,2)	NOT NULL DEFAULT 0.35	Layer fill opacity (0–1)
created_by	UUID	FK → USERS	Officer who created the overlay
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp

4.11 GIS_LAYERS

All map layers available in the system layer panel.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Layer identifier

name	VARCHAR(100)	NOT NULL	Display name
description	TEXT	NULL	Layer description for the UI
type	VARCHAR(30)	NOT NULL	wms | wfs | geojson | raster | eo_satellite
source_url	TEXT	NOT NULL	GeoServer WMS/WFS endpoint or tile URL
style_name	VARCHAR(100)	NULL	GeoServer SLD style name
colour_code	VARCHAR(7)	NULL	Fallback colour for the layer legend dot
access_level	VARCHAR(20)	NOT NULL	public | authenticated | staff | eo_only
is_visible_default	BOOLEAN	NOT NULL DEFAULT FALSE	Whether the layer is on by default
display_order	INTEGER	NOT NULL	Sort order in the layer panel
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp

4.12 PAYMENTS

Fee payments recorded against development applications.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Payment identifier
application_id	VARCHAR(20)	FK → DEVELOPMENT_APPLICATIONS	Associated application
amount	NUMERIC(10,2)	NOT NULL	Payment amount in USD
payment_date	DATE	NOT NULL	Date payment was received
payment_method	VARCHAR(30)	NOT NULL	cash | bank_transfer | mobile_money
reference	VARCHAR(50)	NOT NULL	Bank reference or receipt number
recorded_by	UUID	FK → USERS	Revenue Clerk who recorded it
created_at	TIMESTAMPTZ	NOT NULL	Record creation timestamp

4.13 NOTIFICATIONS

System messages sent to users.

Attribute	Data Type	Constraint	Description
id	SERIAL	PK	Notification identifier
user_id	UUID	FK → USERS	Recipient user
type	VARCHAR(50)	NOT NULL	application_submitted | approved | rejected | inspection_scheduled | payment_due | system
title	VARCHAR(200)	NOT NULL	Short notification title
message	TEXT	NOT NULL	Notification body text
reference_id	VARCHAR(30)	NULL	Related record ID (application, proposal, etc.)
is_read	BOOLEAN	NOT NULL DEFAULT FALSE	Whether user has read the notification
created_at	TIMESTAMPTZ	NOT NULL	Timestamp notification was generated

4.14 AUDIT_LOGS

Immutable record of all system actions.

Attribute	Data Type	Constraint	Description
id	BIGSERIAL	PK	Log entry identifier
user_id	UUID	FK → USERS	User who performed the action
action	VARCHAR(20)	NOT NULL	CREATE | UPDATE | DELETE | LOGIN | LOGOUT | VIEW
table_name	VARCHAR(50)	NOT NULL	Database table affected
record_id	VARCHAR(50)	NULL	Primary key of the affected record
old_values	JSONB	NULL	Previous field values (UPDATE/DELETE)
new_values	JSONB	NULL	New field values (CREATE/UPDATE)
ip_address	VARCHAR(45)	NULL	Client IP address
timestamp	TIMESTAMPTZ	NOT NULL	Exact timestamp of the action

5. PHYSICAL DATA MODEL

The physical model specifies how the logical entities are implemented in the PostgreSQL + PostGIS database. All geometry columns use the WGS84 coordinate reference system (SRID 4326). Spatial indexes use the GiST (Generalized Search Tree) method for efficient bounding box and nearest-neighbour queries.

5.1 Indexes

Table	Index	Purpose
users	idx_users_email	B-tree on email — fast login lookup
users	idx_users_role_id	B-tree on role_id — role filtering
wards	idx_wards_geometry	GiST on geometry — spatial queries
stands	idx_stands_ward_id	B-tree on ward_id — ward filtering
stands	idx_stands_geometry	GiST on geometry — bounding box queries
stands	idx_stands_zone_id	B-tree on zone_id — zone filtering
development_applications	idx_da_applicant_id	B-tree on applicant_id — user application list
development_applications	idx_da_stand_id	B-tree on stand_id — stand application history
development_applications	idx_da_status	B-tree on status — status filtering
proposals	idx_proposals_ward_id	B-tree on ward_id — ward proposals
proposals	idx_proposals_stage	B-tree on stage — stage filtering
proposals	idx_proposals_geometry	GiST on geometry — spatial overlap queries
environmental_overlays	idx_env_overlay_geometry	GiST on geometry — conflict checks
audit_logs	idx_audit_timestamp	B-tree on timestamp — log chronological queries
audit_logs	idx_audit_user_id	B-tree on user_id — per-user audit trail
payments	idx_payments_application_id	B-tree on application_id — fee balance queries

5.2 Constraints and Business Rules

- A user's email must be globally unique across the users table.
- A stand can have at most one active development application at a time (status not in (approved, rejected, withdrawn)).
- A proposal's geometry must fall within the boundary of its assigned ward (enforced via PostGIS ST_Within trigger).
- Payment amounts must be positive (CHECK amount > 0).
- application fee_paid must never exceed fee_required (enforced in the Fastify API layer).
- Audit log records are append-only — no UPDATE or DELETE is permitted on the audit_logs table.
- Role assignments can only be made by users with the admin or hr_admin role.
- Environmental overlay geometries must be valid PostGIS geometries (ST_IsValid enforced at insert).
- All timestamps are stored in UTC (TIMESTAMPTZ) and converted to Africa/Harare (UTC+2) in the API response.

6. APPLICATION WINDOWS AND SCREENS

This section catalogues every screen in the frontend application. Each screen entry describes its purpose, primary users, and key functions. A blank layout space is provided after each entry for wireframes or screenshots to be inserted. **1 Design Principles**

Principle	Implementation
Map-centric	Maplibrie.js interactive map is the primary workspace; all ward, stand, and layout data visible on map
Role-aware	Navigation menus, dashboard tiles, available tools, and visible GIS layers change based on role (admin/eo/planner/user)
Responsive	Full functionality on desktop; simplified view on tablet for field-use by Building Inspectors
Accessibility	WCAG 2.1 AA; keyboard navigation; sufficient colour contrast
Bilingual-ready	Interface designed to accommodate both English and Shona/Ndebele label additions
Low-bandwidth friendly	GeoServer tile caching minimises data transfer for rural areas with limited connectivity

6.2 Screen Descriptions

9.2.1 /development-application — Public Permit Submission

The public application screen is accessible without login. It is the primary interface for community members submitting permit requests to Vungu RDC.

Vungu RDC Portal
MUNICIPAL GIS DIVISION

Home | Submit | Track | Guidelines | Contact | Sign In

Submit a Development Application

Official portal for land development and building permits within the Vungu Rural District Council jurisdiction.

Application Details

Application Type

Select application type...

Full Name

e.g. John Doe

Phone Number

+263 ...

Email Address

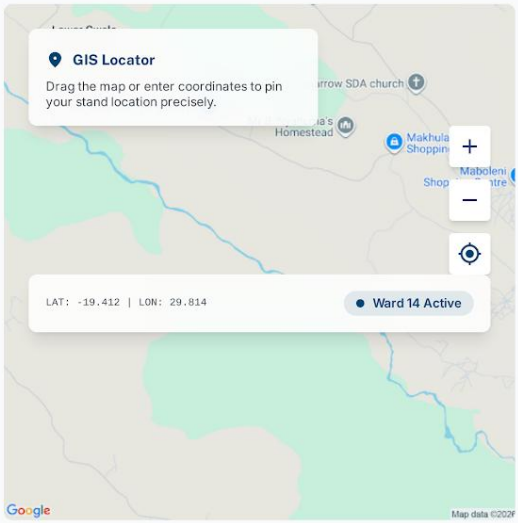
applicant@example.com

Ward Selection

Select ward...

Stand Reference (Optional)

e.g. ST-2044-B



Submit Application

Track Your Application

Already submitted? Check your status using your reference number.

REF-0000-000

Find

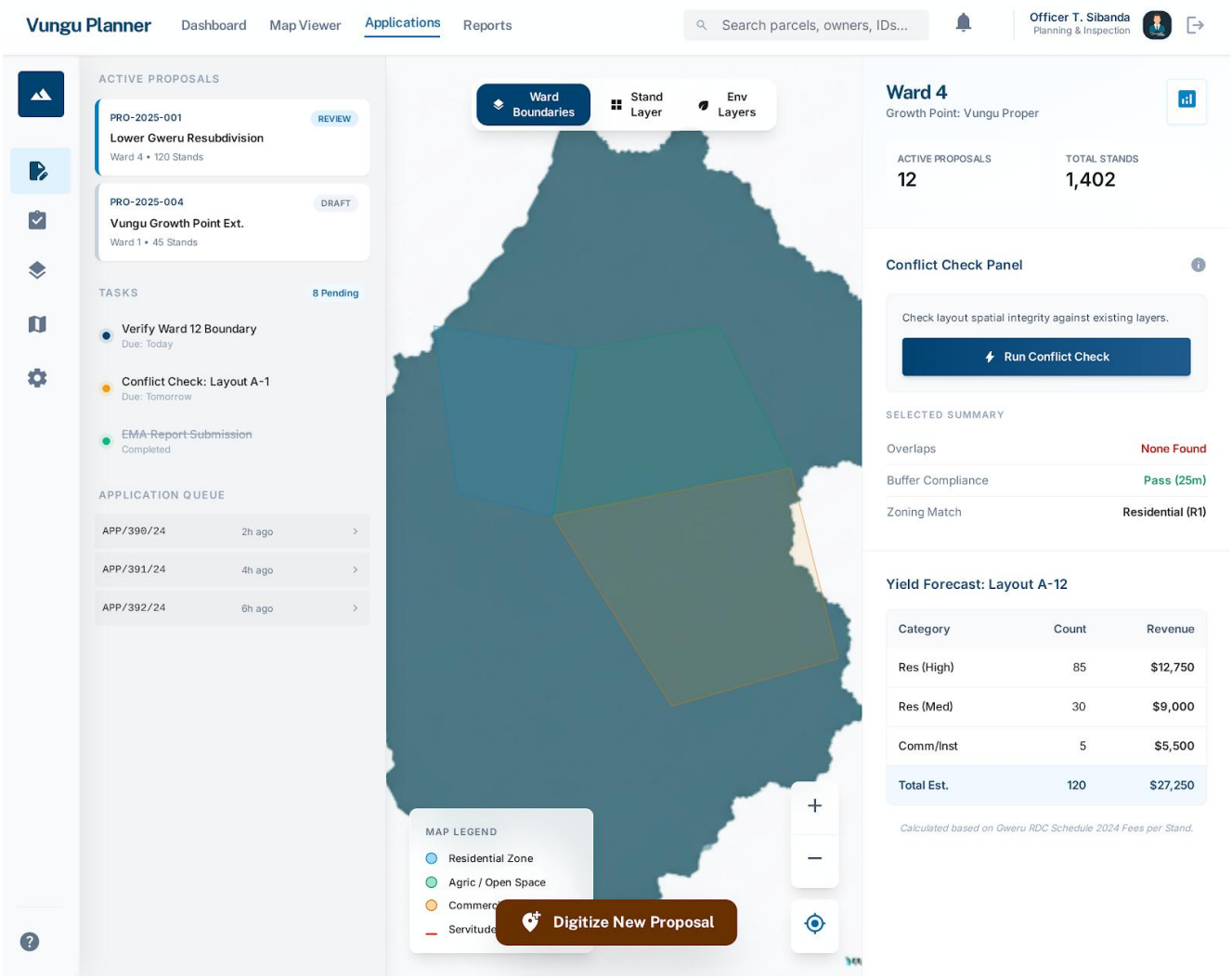
Standard Processing
Building permits are processed within 14 working days of submission.

Legal Requirements
All submissions must comply with the Rural District Councils Act [Chapter 29:13].

Need Assistance?
Contact our GIS helpdesk at support@vungurdc.org.zw

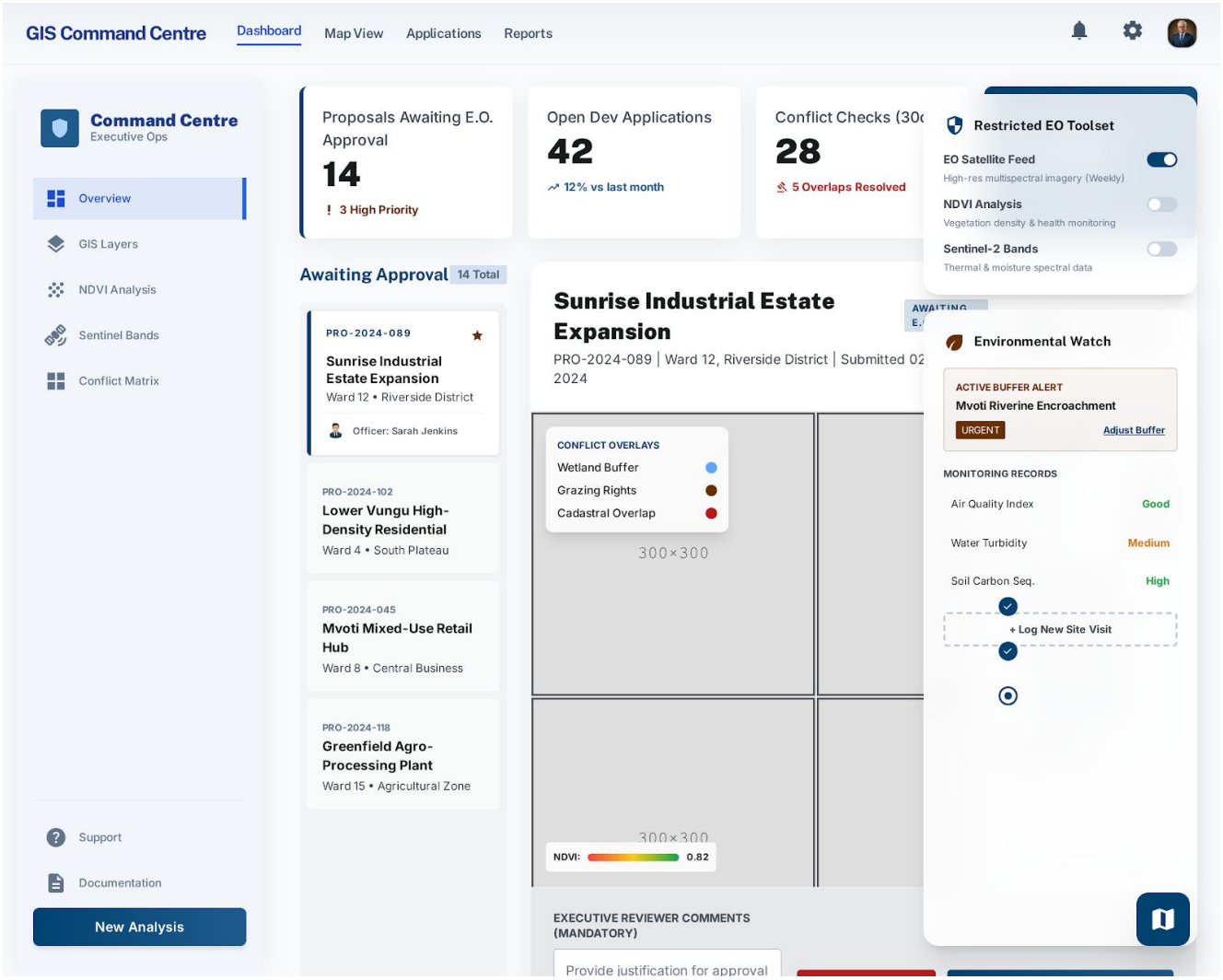
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6.2.2 /planner — Planning Officer Workstation

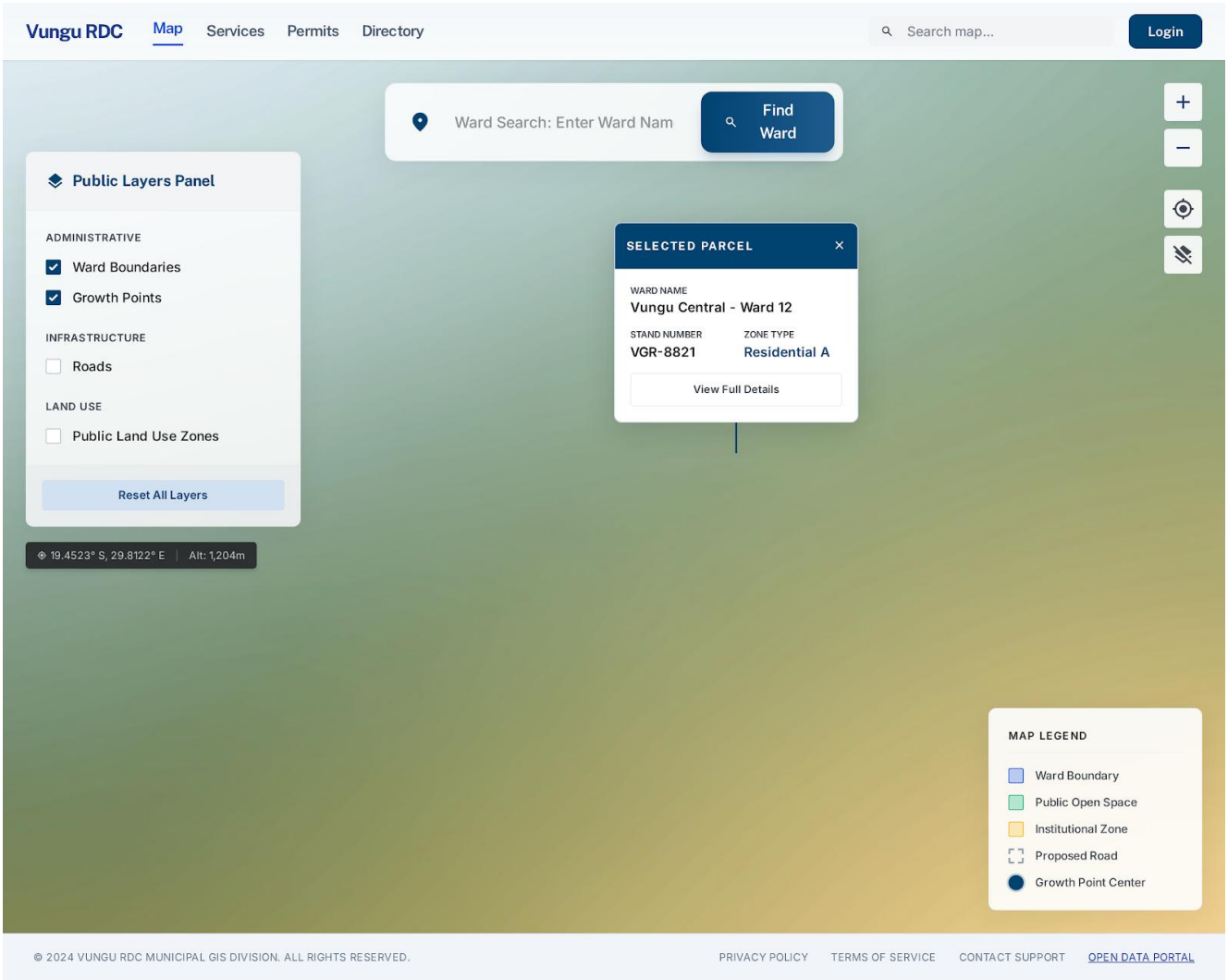


6.2.3 /eo-planner — E.O. Command Centre

The /eo-planner is the executive interface for the E.O. Planning & Environment and the Environment Officer.


GIS Command Centre
Dashboard
Map View
Applications
Reports
Command Centre
Executive Ops
Overview
GIS Layers
NDVI Analysis
Sentinel Bands
Conflict Matrix
Support
Documentation
New Analysis
Proposals Awaiting E.O. Approval
14
3 High Priority
Open Dev Applications
42
12% vs last month
Conflict Checks (30
28
5 Overlaps Resolved
Restricted EO Toolset
EO Satellite Feed
High-res multispectral imagery (Weekly)
NDVI Analysis
Vegetation density & health monitoring
Sentinel-2 Bands
Thermal & moisture spectral data
Awaiting Approval
14 Total
PRO-2024-089
Sunrise Industrial Estate Expansion
Ward 12 • Riverside District
Officer: Sarah Jenkins
PRO-2024-102
Lower Vungu High-Density Residential
Ward 4 • South Plateau
PRO-2024-045
Mvoti Mixed-Use Retail Hub
Ward 8 • Central Business
PRO-2024-118
Greenfield Agro-Processing Plant
Ward 15 • Agricultural Zone
Sunrise Industrial Estate Expansion
PRO-2024-089 | Ward 12, Riverside District | Submitted 02
2024
CONFLICT OVERLAYS
Wetland Buffer
Grazing Rights
Cadastral Overlap
300×300
300×300
NDVI:
0.82
Environmental Watch
ACTIVE BUFFER ALERT
Mvoti Riverine Encroachment
URGENT
Adjust Buffer
MONITORING RECORDS
Air Quality Index
Good
Water Turbidity
Medium
Soil Carbon Seq.
High
+ Log New Site Visit
EXECUTIVE REVIEWER COMMENTS (MANDATORY)
Provide justification for approval

6.2.4 /explorer — Public GIS Viewer



7. REPORTS — FORMAT AND FIELD SPECIFICATIONS

This section defines every report generated by the system: who generates it, when, in what format it is delivered, and what fields it contains.

7.1 Development Application Status Report

Property	Value
Generated By	Planning Officer, E.O. Planning and Environment
Trigger	On demand; also auto-generated monthly on the 1st of each month
Format	On-screen HTML table + downloadable PDF / CSV
Filters	Date range, ward, application type, status, assigned officer
Scope	All development applications within the selected filters

Field / Column	Type	Description
Application Reference	String	Unique application ID (e.g. DA-2026-0001)
Applicant Name	String	Full name of the submitting applicant
Stand Number	String	Stand number subject to the application
Ward	String	Ward name and number
Application Type	String	new_build | extension | change_of_use | subdivision | demolition
Submitted Date	Date	Date the application was submitted
Current Status	String	submitted | under_review | awaiting_inspection | approved | rejected
Days in Current Status	Integer	Number of calendar days since last status change
Decision Date	Date	Date the final decision was made (if applicable)
Decision Officer	String	Name of officer who made the decision
Fee Required	Currency	Total application fee due (USD)
Fee Paid	Currency	Total amount paid to date (USD)
Outstanding Balance	Currency	fee_required minus fee_paid

7.2 Proposal Approval Summary Report

Property	Value
Generated By	E.O. Planning and Environment, Planning Officer
Trigger	On demand; monthly summary auto-generated
Format	On-screen table + downloadable PDF
Scope	All proposals within the selected period and ward filter

Field / Column	Type	Description
Proposal Reference	String	Unique proposal ID (e.g. PRO-2026-001)
Title	String	Proposal descriptive title
Ward	String	Ward name
Prepared By	String	Planning Officer full name

Date Created	Date	Proposal creation date
Current Stage	String	draft | circulating | awaiting_eo | approved | rejected | billed
Approved / Rejected By	String	E.O. who made the decision
Decision Date	Date	Date approved or rejected
Total Stands	Integer	Total planned stands in the layout
Total Area (ha)	Decimal	Drawn layout area in hectares
Estimated Revenue	Currency	Sum of application fees across all planned stands (USD)
Decision Notes	Text	E.O.'s written rationale for approval or rejection

7.3 Ward Yield Analysis Report

Property	Value
Generated By	Planning Officer
Trigger	Auto-generated when a planning polygon is closed on the map; also on demand
Format	On-screen table in the Yield tab of the Vungu Planner + downloadable CSV
Scope	A single drawn polygon on the map (layout area)

Field / Column	Type	Description
Proposal Reference	String	Associated proposal ID
Ward	String	Ward name
Total Layout Area (ha)	Decimal	Polygon area calculated by Turf.js
Stand Type	String	Residential | Commercial | Industrial | Open Space
Land Use Percentage (%)	Integer	Percentage of total area allocated to this stand type
Estimated Stand Count	Integer	Number of stands derivable from area and minimum plot size
Average Plot Size (m²)	Integer	Average stand size for this type
Application Fee / Stand	Currency	Statutory fee per stand for this type (USD)
Estimated Revenue	Currency	Stand count × application fee for this type (USD)
% of Total Revenue	Decimal	This type's revenue as a percentage of total
TOTAL	Row	Sum row: total stand count and total estimated revenue

7.4 Conflict Check Report

Property	Value
Generated By	Planning Officer, E.O. Planning and Environment, Environment Officer
Trigger	On demand when the user clicks "Run Conflict Check" in the Planner
Format	On-screen panel with colour-coded result (green/amber/red) + downloadable PDF
Scope	Currently drawn polygon or selected feature against all active environmental overlays

Field / Column	Type	Description

Check Reference	String	Auto-generated reference (CHK-2026-XXXXX)
Proposal / Geometry Ref	String	Proposal ID or drawn polygon identifier
Check Date	DateTime	Timestamp when the check was run
Run By	String	Officer who ran the check
Overall Result	String	CLEAR | WARNING | CONFLICT
Overlay Name	String	Environmental overlay that intersected (if any)
Overlay Type	String	wetland | river_buffer | road_reserve | protected_area
Intersection Area (m²)	Decimal	Area of overlap between the geometry and the overlay
Buffer Distance (m)	Decimal	Required buffer distance from the constraint
Encroachment (m)	Decimal	Distance by which the geometry exceeds the buffer
Recommended Action	Text	System-generated guidance (e.g. "Adjust north-eastern edge by 20 m")

7.5 Inspection Report

Property	Value
Generated By	Building Inspector
Trigger	Generated when the inspector submits a completed inspection record
Format	On-screen view + printable PDF (A4, formal letterhead)
Scope	A single on-site inspection visit

Field / Column	Type	Description
Inspection Reference	String	Unique inspection ID (INS-2026-XXXXX)
Application Reference	String	Associated development application
Stand Number	String	Stand number inspected
Ward	String	Ward name
Applicant Name	String	Property owner / applicant
Inspector Name	String	Building Inspector full name
Inspection Stage	String	foundation | slab | walling | roofing | completion
Scheduled Date	Date	Planned inspection date
Conducted Date	Date	Actual inspection date
Result	String	pass | fail | conditional_pass | not_conducted
Observations	Text	Inspector's field observations and measurements
Defects / Non-Compliance	Text	List of identified defects with required remediation
Next Inspection Stage	String	Next required inspection (if applicable)
Inspector Signature	Image	Digital or hand-signed signature block
Photos	Image[]	Attached inspection photographs

7.6 Monthly Revenue Collection Report

Property	Value
Generated By	Revenue Clerk, E.O. Income, Council Treasurer
Trigger	Auto-generated on the 1st of each month for the previous month
Format	On-screen table + downloadable PDF / Excel
Scope	All payments recorded in the selected month

Field / Column	Type	Description
Report Period	String	Month and year (e.g. March 2026)
Payment Reference	String	Bank reference or receipt number
Application Reference	String	Associated development application
Stand Number	String	Stand subject to the application
Ward	String	Ward name
Applicant Name	String	Payer full name
Application Type	String	new_build | extension | change_of_use | etc.
Amount Paid	Currency	Payment amount (USD)
Payment Date	Date	Date payment was received
Payment Method	String	cash | bank_transfer | mobile_money
Recorded By	String	Revenue Clerk who recorded the payment
Running Total	Currency	Cumulative total for the month to this row
TOTAL COLLECTED	Row	Grand total of all payments in the period
Total Applications	Integer	Count of distinct applications with payments this period
Outstanding (all apps)	Currency	Total outstanding fees across all active applications

7.7 Environmental Compliance Report

Property	Value
Generated By	Environment Officer, E.O. Planning and Environment
Trigger	Quarterly; also on demand
Format	On-screen view + downloadable PDF including embedded map snapshot
Scope	All development applications and proposals within the reporting period

Field / Column	Type	Description
Report Period	String	Quarter and year (e.g. Q1 2026)
Ward	String	Ward name (one row per ward)
Total Applications Reviewed	Integer	Total applications reviewed for environmental compliance
Applications Cleared	Integer	Applications with no environmental conflicts
Applications with Warnings	Integer	Applications encroaching within buffer zones
Applications with Hard Conflicts	Integer	Applications overlapping protected areas or road reserves
Most Common Conflict Type	String	Overlay type most frequently triggered
Total Buffer Encroachment Area (m²)	Decimal	Total area of buffer zone encroachments detected
Proposals Rejected (Env. Grounds)	Integer	Proposals rejected by E.O. on environmental grounds
NDVI Change (%)	Decimal	Vegetation index change in the ward year-on-year (from satellite data)
Overlay Updates This Period	Integer	Number of environmental overlay modifications
Officer Notes	Text	Environment Officer's observations and recommendations

7.8 CEO Executive Summary Report

Property	Value
Generated By	System (auto-generated); distributed to CEO
Trigger	Auto-generated monthly on the 1st; available in the CEO dashboard
Format	Dashboard view with charts and KPI cards + downloadable PDF (formal report format)
Scope	All 20 wards, all departments, for the preceding month

Field / Column	Type	Description
Report Month	String	Month and year
Total Applications Received	Integer	New applications submitted this month
Total Applications Approved	Integer	Applications approved this month
Total Applications Rejected	Integer	Applications rejected this month
Total Applications Pending	Integer	Applications currently under review
Average Processing Time (days)	Decimal	Average days from submission to decision
Revenue Collected (USD)	Currency	Total development fees collected this month
Outstanding Revenue (USD)	Currency	Total outstanding fees across all active applications
Proposals Approved	Integer	Planning proposals approved by E.O. this month
Inspections Conducted	Integer	Total building inspections this month
Inspection Pass Rate (%)	Decimal	Percentage of inspections that passed
Most Active Ward	String	Ward with highest application volume
Environmental Conflicts Detected	Integer	Total conflict check warnings/errors issued
New Registered Users	Integer	Public users who registered this month
System Uptime (%)	Decimal	Frontend/backend availability percentage

7.9 Audit Trail Report

Property	Value
Generated By	Auditor, Audit Clerk, System Administrator
Trigger	On demand; filters by user, date range, action type, or table
Format	On-screen paginated table + downloadable CSV / PDF
Scope	All create/update/delete actions within the selected filter criteria

Field / Column	Type	Description
Log ID	Integer	Unique audit log entry identifier
Timestamp	DateTime	Exact date and time of the action (UTC+2)
User	String	Full name and role of the user who acted
Action	String	CREATE | UPDATE | DELETE | LOGIN | LOGOUT | VIEW
Module / Table	String	System module or database table affected
Record ID	String	Primary key of the affected record

Previous Values	JSON	Field values before the change (UPDATE/DELETE only)
New Values	JSON	Field values after the change (CREATE/UPDATE only)
IP Address	String	Client IP address at time of action
Session ID	String	Browser session identifier

8. NAVIGATION FLOW AND USER WORKFLOWS

8.1 Authentication Flow

Step	Action	System Behaviour
1	User visits the system URL	Vue Router checks if a valid JWT token exists in localStorage
2	No token present	User is redirected to the Login page (/login)
3	User submits credentials	Frontend sends POST /api/auth/login with email and password
4	Backend validates	Fastify verifies bcrypt hash; if valid, issues HMAC-SHA256 JWT
5	JWT stored	Token stored in localStorage; Pinia auth store updated
6	Role-based redirect	Vue Router reads user.role and redirects to the role's landing_path
7	On each page load	initializeAuth() restores session from localStorage on app startup
8	Token expiry / logout	Auth store is cleared; user redirected to /login with session-expired message

8.2 Development Application Workflow

Step	Actor	Action
1	Public User	Submits application via the 5-step form (/development-application)
2	System	Generates application reference (DA-YYYY-XXXX); sets status to submitted; sends confirmation notification to applicant
3	Planning Officer	Receives notification; opens application in the Application Review queue
4	Planning Officer	Reviews plans and description; runs spatial conflict check; changes status to under_review
5	Planning Officer	Schedules a building inspection (status: awaiting_inspection)
6	Building Inspector	Conducts on-site inspection; records result in the system
7	Planning Officer	Reviews inspection result; prepares decision recommendation
8	E.O. Planning	Reviews recommendation; approves or rejects application; records decision notes
9	System	Updates application status; sends notification to applicant with decision and any conditions
10	Revenue Clerk	Records application fee payment when received (if approved)
11	System	Updates fee_paid; marks application as paid when balance is cleared

8.3 Planning Proposal Workflow

Step	Actor	Action
1	Planning Officer	Opens Vungu Planner; draws a layout polygon on the map for a specific ward
2	System (Turf.js)	Automatically calculates area, perimeter, and opens the Yield Table in the right sidebar
3	Planning Officer	Reviews yield table; adjusts layout polygon as needed; runs

		conflict check
4	Planning Officer	Saves the proposal (status: draft) with title, ward, notes, and geometry
5	Planning Officer	Submits proposal for circulation (status: circulating) — circulates to Roads Engineer and other departments for comment
6	Roads Engineer / Dept Officers	Reviews and adds comments in the Comments tab; flags any road reserve conflicts
7	Planning Officer	Addresses comments; advances to awaiting_eo (status: awaiting_eo)
8	E.O. Planning	Reviews proposal in the E.O. Command Centre; runs final conflict check on environmental overlays
9	E.O. Planning	Approves or rejects the proposal (status: approved / rejected); records decision notes
10	Planning Officer	On approval: prepares final layout plan documents; links to application billing
11	Council Treasurer / Revenue Clerk	Bills applicants for application fees based on the approved yield table (status: billed)

9. SYSTEM PERMISSIONS MATRIX

The table below defines which roles have access to each major system module. R = Read, W = Write/Create, U = Update, D = Delete, — = No Access.

Module	Admin	CEO	E.O.	Planner	Inspector	Env. Officer	Treasurer	Revenue Clerk	Auditor	HR Admin	Roads Eng.	Public
Home / Public Map	R	R	R	R	R	R	R	R	R	R	R	R
Apply for Development Permit (public form)	—	—	—	—	—	—	—	—	—	—	—	W
Application Status Tracker (own applications)	—	—	—	—	—	—	—	—	—	—	—	R
Vungu Planner (full GIS workstation)	—	—	R	RWU	RWU	R	—	—	—	—	R	—
E.O. Command Centre	—	—	RWU	—	—	RWU	—	—	—	—	—	—
Approve / Reject Proposals (E.O. decision)	—	—	U	—	—	—	—	—	—	—	—	—
Approve / Reject Applications (officer decision)	—	—	U	U	—	—	—	—	—	—	—	—
Create & Submit Planning Proposals	—	—	—	W	—	—	—	—	—	—	—	—
Inspection Records (create & update)	—	—	R	R	RWU	—	—	—	R	—	—	—
Environmental Overlays (create & update)	—	—	RWU	—	—	RWU	—	—	R	—	—	—
Yield Analysis Tool	—	—	R	RW	—	—	R	—	R	—	—	—
Conflict Check Tool	—	—	RW	RW	—	RW	—	—	R	—	—	—
Finance Dashboard	R	R	—	—	—	—	R	R	R	—	—	—
Record Payments	—	—	—	—	—	—	R	RW	—	—	—	—
Debtor Management	—	—	—	—	—	—	R	RWU	R	—	—	—
Analytics Dashboard	R	R	R	R	R	R	R	—	R	R	R	—
User Management (create / edit users)	RWU	—	—	—	—	—	—	—	—	RWU	—	—
Role Assignment	RWU	—	—	—	—	—	—	—	—	RWU	—	—
Audit Log Viewer	R	—	—	—	—	—	—	—	R	—	—	—
CEO Executive Dashboard	R	R	—	—	—	—	R	—	—	—	—	—
GIS Layer Management (admin)	RWU	—	—	—	—	—	—	—	—	—	—	—

Key: R = Read only W = Create/Write U = Update RW = Read + Create RWU = Full access — = No access

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